JFET Preamp Input Impedance Analysis

JFET Impedance characterization — 1 MHz to 100 MHz

Devices: fmpre0, fmpre2, fmpre5, fmpre7 | n = 10 sweeps each

Measurement by Emi Tamura, Analyzed by Aritoki Suzuki – June 9th 2026

Measurement Overview

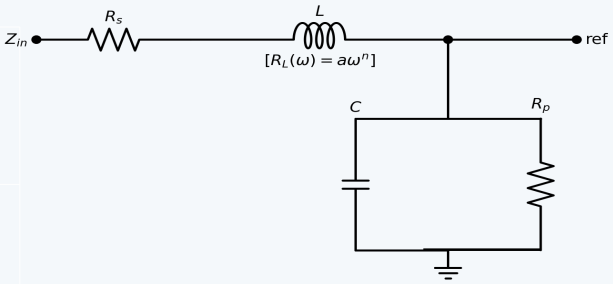
4 JFET preamps measured
fmpre0, 2, 5, 7 — 10 impedance sweeps each

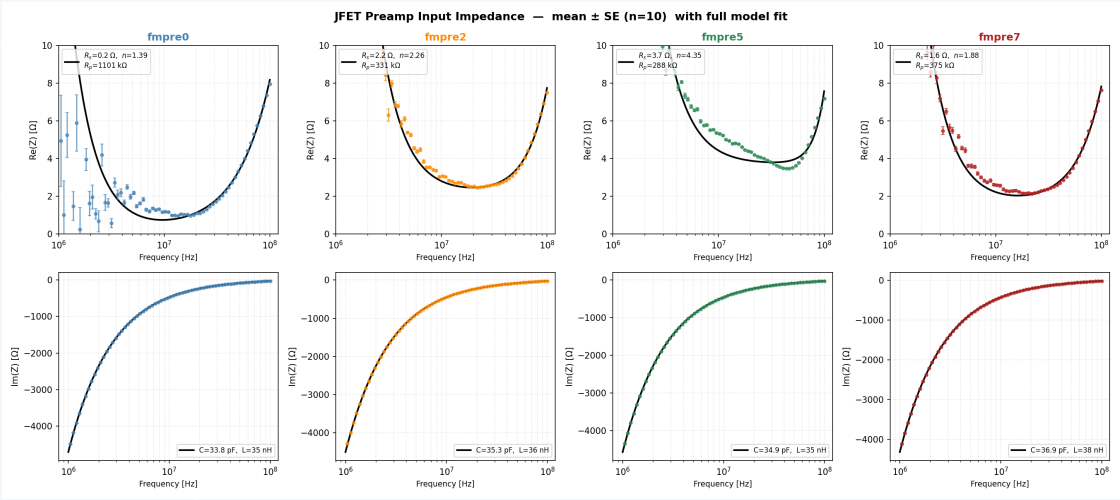
Frequency range
100 Hz – 100 MHz (201 log-spaced points)

Useful range for modelling
1 MHz – 100 MHz (below 1 MHz: noisy)

Both components extracted
Re(Z) and Im(Z) vs frequency

Analysis goal
Find equivalent circuit + fit parameters





Re(Z) top row, Im(Z) bottom row — mean ± SE overlaid with fitted model

Im(Z) — Series L–C Model

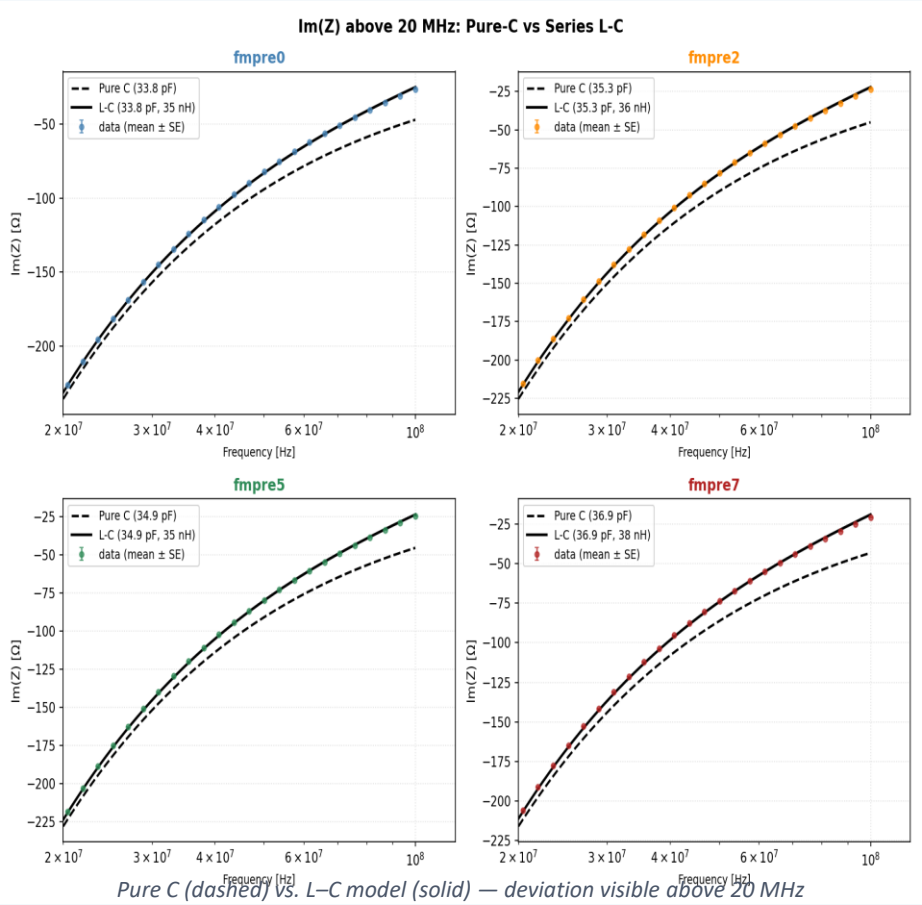
$$\mathrm{Im}(Z) = \omega L - 1/(\omega C)$$

- **Below ~20 MHz → pure capacitor**
  Im ≈ −1/(ωC): apparent C is nearly constant ~34–37 pF
  → dominated by JFET gate capacitance (C_gs + C_gd)

- **Above ~20 MHz → inductance kicks in**
  Series inductance L pulls Im(Z) back toward zero
  (magnitude falls faster than 1/ωC)

- **L ≈ 35–39 nH**
  Corresponds to ~35 mm PCB trace from device to measurement point

- **Fit quality**
  L–C model fit much better across all frequency range



Pure C (dashed) vs. L–C model (solid) — deviation visible above 20 MHz

Re(Z) — Two Competing Components

Gate leakage (falls with f)

$$1 / (\omega^2 R_p C^2)$$

Parallel resistance R_p (JFET gate reverse-bias leakage) shunts the capacitor. Contribution to Re(Z) falls steeply as $1/f^2$ — dominant at 5–20 MHz. R_p > 288 kΩ across devices.

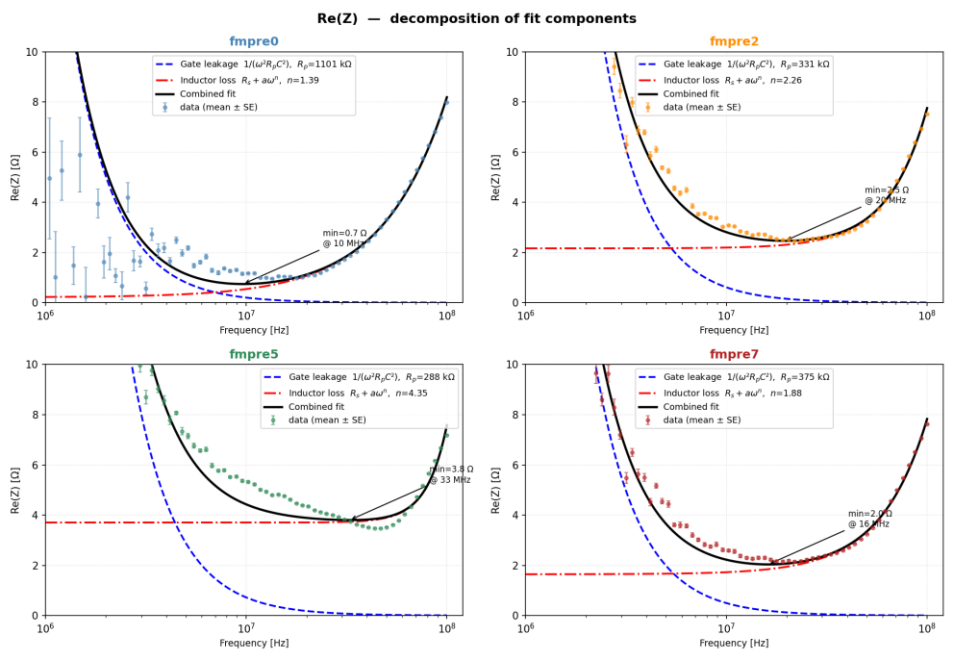
+

Inductor loss (rises with f)

$$R_s + a \cdot \omega^n$$

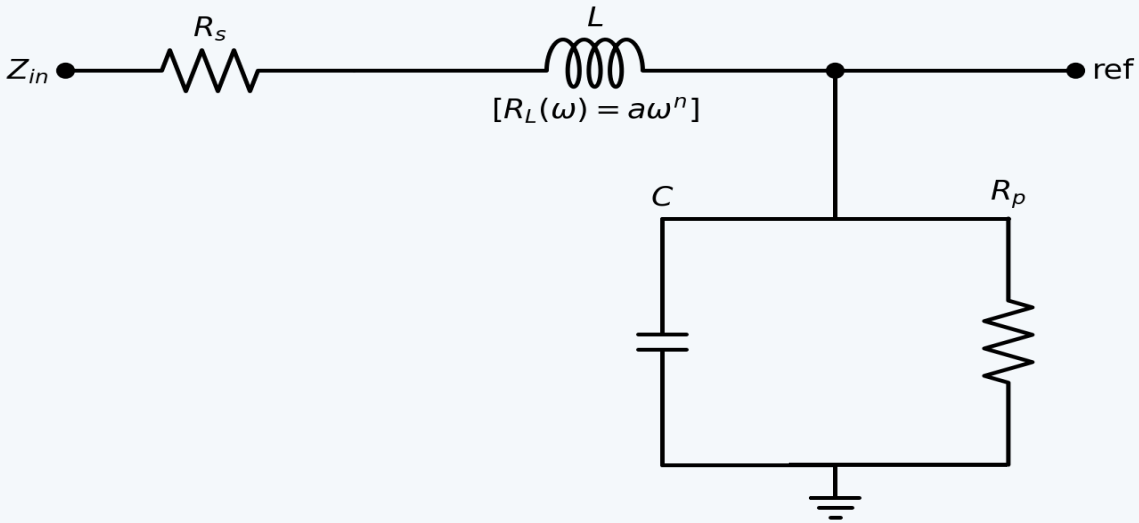
Frequency-dependent series resistance of the PCB trace (skin & proximity effect). Rises as ω^n ($n \approx 1.4$–4.4). Dominates above ~30 MHz; reaches 6–8 Ω at 100 MHz.

Combined → U-shaped Re(Z)



Blue dashed = gate leakage | Red dash-dot = inductor loss | Black = combined fit

Full Equivalent Circuit Model



Series R_s + R_L(ω)

DC series resistance + frequency-dependent inductor loss (skin effect). Accounts for rising Re(Z) above 30 MHz.

Inductance L

PCB trace inductance (~35–39 nH). Causes Im(Z) to deviate from pure −1/ωC above 20 MHz.

C || R_p (shunt to GND)

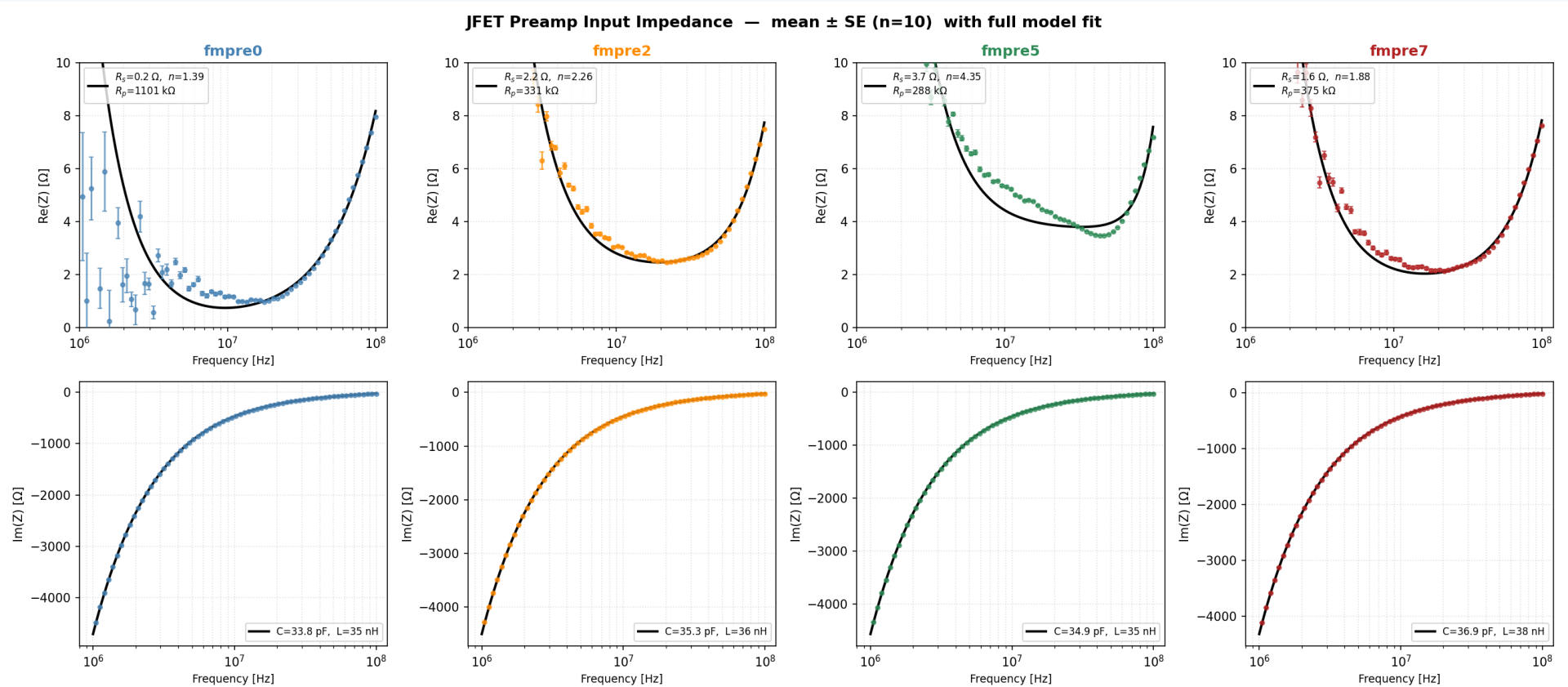
Gate capacitance C (34–37 pF) in parallel with gate leakage R_p (> 288 kΩ). Dominates Im(Z) and low-freq Re(Z).

Fitted Parameters — All 4 Preamps

Preamp	C (pF)	L (nH)	R_s (Ω)	n	a (Ω·s^n)	R_p (kΩ)
fmpre0	33.79 ± 0.03	35.0 ± 1.8	0.21 ± 0.01	1.39 ± 0.00	5.2×10^{-12}	1101 ± 18
fmpre2	35.32 ± 0.04	36.4 ± 2.0	2.16 ± 0.00	2.26 ± 0.01	6.9×10^{-20}	331 ± 2
fmpre5	34.89 ± 0.04	34.7 ± 1.9	3.71 ± 0.00	4.35 ± 0.01	2.2×10^{-38}	288 ± 1
fmpre7	36.85 ± 0.05	38.5 ± 2.5	1.65 ± 0.01	1.88 ± 0.01	1.9×10^{-16}	375 ± 2

Note on a: a is the coefficient in $R_L(\omega) = a\cdot\omega^n$, with units Ω·$s^n$. Its value alone is not directly interpretable — pair it with n. To get the actual inductor resistance at any frequency: $R_L(f) = a\cdot(2\pi f)^n$. Example: at 100 MHz, fmpre0 gives $R_L \approx 8.0$ Ω.

Fitted Parameters — All 4 Preamps



Physical Interpretation — JFET Context

C **Gate Capacitance C ≈ 34–37 pF**

Physically: C_gs + C_gd of the JFET. Device-to-device spread (34–37 pF) reflects manufacturing variation. This is the dominant input capacitance and sets the noise bandwidth of the preamp.

L **Series Inductance L ≈ 35–39 nH**

This inductance is not the JFET itself — it arises from the PCB trace (~35 mm) between the device and the measurement probe. Important to de-embed when comparing across setups. Bond wire alone contributes ~1–2 nH.

R_p **Gate Leakage R_p > 288 kΩ**

Parallel resistance from the gate reverse-bias leakage current. All four values are safely high (≫ 1 kΩ), confirming negligible DC leakage noise contribution at operating frequencies.

R_L **Inductor Loss R_L(ω) — power law ω^n**

Skin and proximity effect in the PCB trace. The varying exponent n (1.3–3) across devices likely reflects real differences in trace geometry or contact resistance at the measurement pads. Does not affect JFET noise directly.

Conclusions

- **Im(Z) → series L–C model** Im(Z) = ωL − 1/(ωC) fits all four preamps well (1–100 MHz). C = 34–37 pF is the gate capacitance; L = 35–39 nH is the PCB trace inductance.

- **Re(Z) → two-component U-shape** The U-shape arises from gate leakage ($R_p \parallel C$, falling as $1/\omega^2$) and inductor skin-effect loss (rising as ω^n). Neither alone suffices.

- **All devices behave as expected JFETs** $R_p \gg 1\ k\Omega$ confirms low gate leakage. Capacitances are consistent and physically reasonable. Unit-to-unit spread is small.

- **PCB parasitics dominate at high frequency** L and R_L(ω) are trace effects, not intrinsic to the JFET. De-embedding the trace inductance would reveal the bare device input impedance.